U.S. Navy A-4D Attack Aircraft

1. State-space

V0 = 583.7849 ft/s, alpha_trim = 8.8 deg

Longitudinal A matrix:

```
  -0.0128   -0.0353         0   -0.0545
  -0.1140   -0.3739    1.0000   -0.0084
   0.2518   -5.3422   -0.6440    0.0013
        0         0    1.0000         0
```

Longitudinal B vector:

```
   0.0046
  -0.0406
  -8.0935
        0
```

Lateral A matrix:

```
  -0.1034         0   -1.0000    0.0545
 -16.3761   -0.6809    0.5420         0
   8.9625    0.0417   -0.3274         0
        0    1.0000    0.1548         0
```

Lateral B matrix:

```
  -0.0008    0.0179
   8.1616    3.6354
  -0.6865   -3.7609
        0         0
```

2. EIGENVALUES, EIGENVECTORS, MODE IDENTIFICATION

--- LONGITUDINAL MODES ---

Short Period: lambda = -0.51187 +2.30537i

Non-dim eigenvector (magnitude, phase):

|Delta_u/V0| = 0.0375, phase = 72.2 deg

|Dalpha| = 1.0000, phase = 0.0 deg

|q*cbar/(2V0)| = 0.0214, phase = 93.2 deg

|Dtheta| = 0.9789, phase = -9.3 deg

Dominant states: Dalpha, q*cbar/(2V0)

Stability: Stable

Phugoid: lambda = -0.00350 +0.08209i

Non-dim eigenvector (magnitude, phase):

|Delta_u/V0| = 0.6585, phase = 97.2 deg

|Dalpha| = 0.0211, phase = 95.4 deg

|q*cbar/(2V0)| = 0.0008, phase = 92.4 deg

|Dtheta| = 1.0000, phase = 0.0 deg

Dominant states: Delta_u/V0, Dtheta

Stability: Stable

--- LATERAL MODES ---

Dutch Roll: lambda = -0.20741 +2.98949i

Non-dim eigenvector (magnitude, phase):

|beta| = 0.6053, phase = -11.7 deg

|p*b/(2V0)| = 0.0770, phase = 93.0 deg

|r*b/(2V0)| = 0.0425, phase = -98.0 deg

|phi| = 1.0000, phase = 0.0 deg

Dominant states: phi, beta

Stability: Stable

Roll: lambda = -0.69982 (real)

Non-dim eigenvector (magnitude, phase):

|beta| = 0.0010

|p*b/(2V0)| = 0.0167

|r*b/(2V0)| = 0.0013

|phi| = 1.0000

Dominant state: phi

Stability: Stable

Spiral: lambda = +0.00291 (real)

Non-dim eigenvector (magnitude, phase):

|beta| = 0.0020

|p*b/(2V0)| = 0.0001

|r*b/(2V0)| = 0.0013

|phi| = 1.0000

Dominant state: phi

Stability: UNSTABLE

--- SUMMARY TABLE ---

Mode	Eigenvalue	Stability
Short Period	-0.51187+2.30537i	Stable
Phugoid	-0.00350+0.08209i	Stable

Dutch Roll	-0.20741+2.98949i	Stable
Roll	-0.69982	Stable
Spiral	+0.00291	UNSTABLE

Longitudinal Modes: The A-4 Skyhawk exhibits two classical longitudinal oscillatory modes. The Short Period mode ($\lambda = -0.5119 \pm 2.3054i,\ \omega_n = 2.36$ rad/s, $\zeta = 0.217,\ T_{1/2} = 1.35$ s) is a well-damped, rapid pitch oscillation dominated by angle of attack and pitch rate, with forward velocity contributing negligibly. This confirms the classical character of the longitudinal mode: a fast aerodynamic restoring response at nearly constant airspeed. The Phugoid ($\lambda = -0.00350 \pm 0.0821i,\ \omega_n = 0.082$ rad/s, $\zeta = 0.043,\ T_{1/2} = 198$ s) is a slow, lightly damped energy-exchange oscillation dominated by $\Delta u/V_0$ and $\Delta\theta$ at nearly equal magnitudes, with angle of attack remaining essentially fixed, which is consistent with the Lanchester interpretation of a gradual climb-dive cycle at constant α. Both modes are stable, with the Short Period providing adequate short-term pitch response and the Phugoid exhibiting the characteristically weak damping typical of this aircraft class.

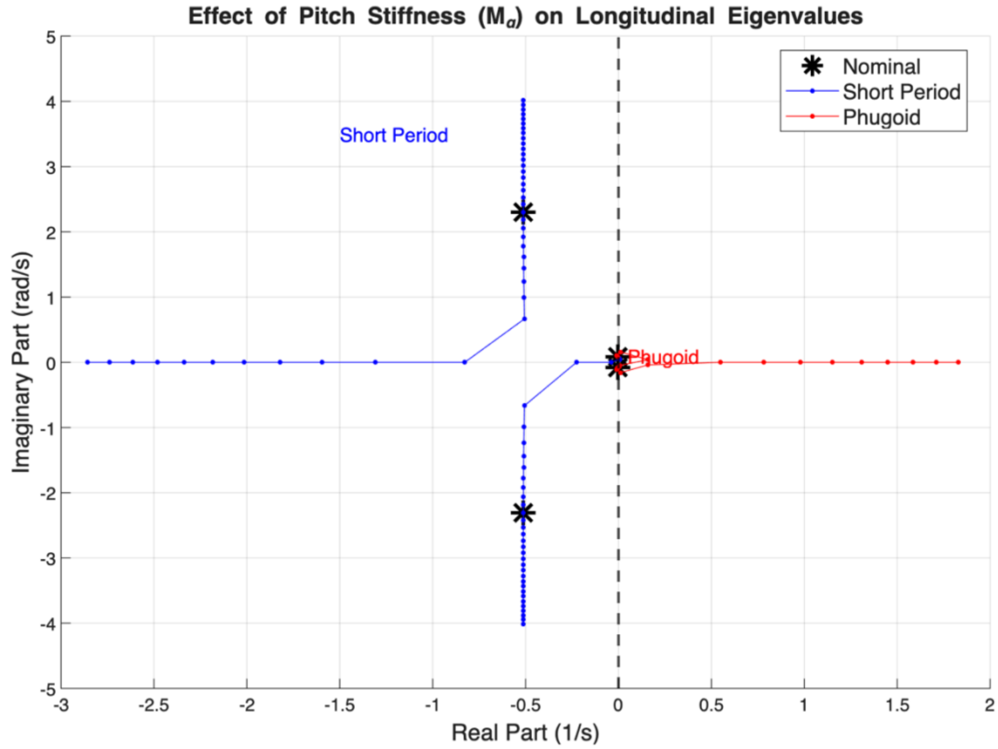
Lateral-Directional Modes: The Dutch Roll ($\lambda = -0.2074 \pm 2.9895i,\ \omega_n = 2.997$ rad/s, $\zeta = 0.069,\ T_{1/2} = 3.34$ s) is a stable but lightly damped oscillatory mode dominated by bank angle and sideslip, representing the familiar coupled roll-yaw weaving motion. The Roll mode ($\lambda = -0.6998,\ \tau = 1.43$ s) is a fast, stable non-oscillatory subsidence governed almost entirely by ϕ, which reflects rapid roll damping with negligible coupling to sideslip or yaw. The Spiral mode ($\lambda = +0.00291,\ T_{double} = 238$ s) is the sole instability, a slow divergence in bank angle that is trivially correctable by the pilot and entirely typical for swept-wing jet aircraft.

3. MODAL PARAMETERS

Mode	wn(r/s)	zeta	wd(r/s)	Period(s)	T_half/dbl(s)
Short Period	2.3615	0.2168	2.3054	2.7255	1.3541 (T_half)
Phugoid	0.0822	0.0426	0.0821	76.5444	198.2576 (T_half)
Dutch Roll	2.9967	0.0692	2.9895	2.1018	3.3420 (T_half)

Mode	Eigenvalue	tau(s)	T_half/dbl(s)
Roll	-0.69982	1.4289	0.9905 (T_half)
Spiral	+0.00291	343.2748	237.9400 (T_double)

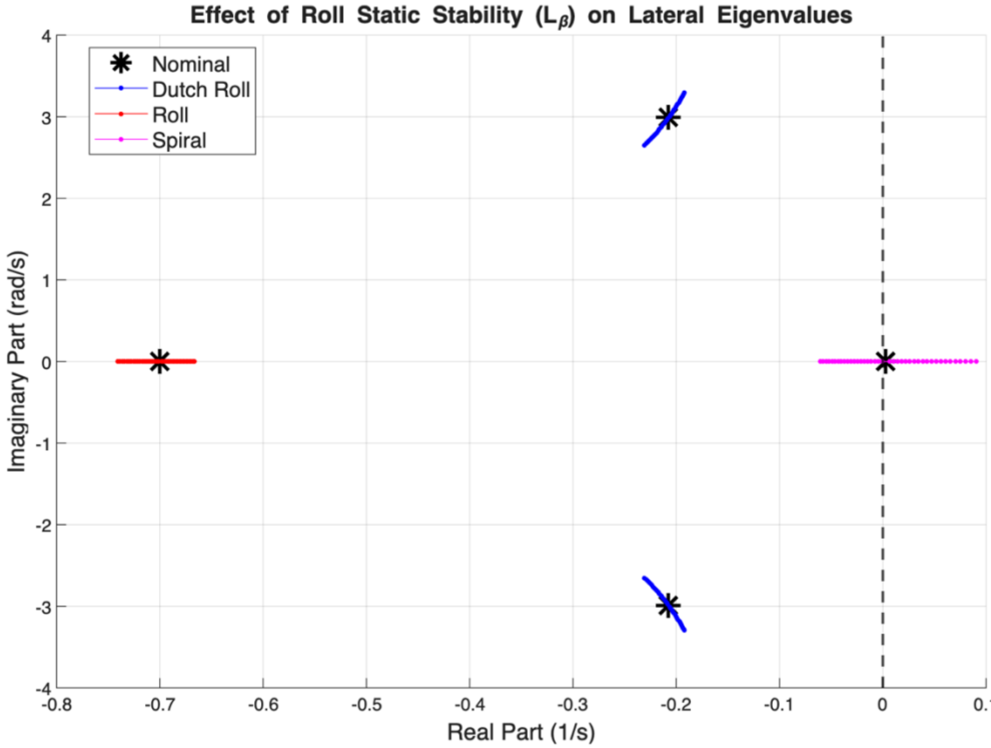
4. PITCH STIFFNESS ROOT LOCUS



Effect of Pitch Stiffness on Longitudinal Eigenvalues. At the nominal condition, the Short Period mode has a natural frequency of ω_n = 2.362 rad/s and a damping ratio of ζ = 0.217, while the Phugoid sits at −0.0035 ± 0.082j (ω_n= 0.082 rad/s, ζ = 0.043). As $|\overline{M_\alpha}|$ is increased up to a factor of 3.0 (k = 20), the Short Period natural frequency rises from 2.362 to 4.048 rad/s while its real part stays nearly fixed at approximately -0.512, causing the damping ratio to decrease from 0.217 to 0.127. The locus therefore sweeps almost vertically upward along a line of nearly constant real part; the mode becomes faster but proportionally less damped with increasing pitch stiffness. On the decreasing branch, as $|\overline{M_\alpha}|$ is reduced toward zero, the Short Period ω_n falls from 2.362 rad/s through 1.695 rad/s at f = 0.5, then down to 0.836 rad/s at f = 0.1 with the damping ratio rising sharply to ζ = 0.606, as the complex pair coalesces toward the real axis and the mode transitions from oscillatory to heavily damped.

When $\overline{M_\alpha}$ goes positive (statically unstable aircraft, f < 0), the coupled system produces a real divergent root, at f = −0.5 a root sits at +1.151 and at f = -1.0 it reaches +1.827, confirming a pitch divergence that would require active stabilization. Throughout the entire sweep, the Phugoid eigenvalue barely moves, drifting only from −0.0035 ± 0.082j at nominal to −0.0026 ± 0.079j at f = 3.0, a negligible change that confirms the slow energy-exchange mode is entirely decoupled from pitch stiffness.

5. ROLL STIFFNESS ROOT LOCUS

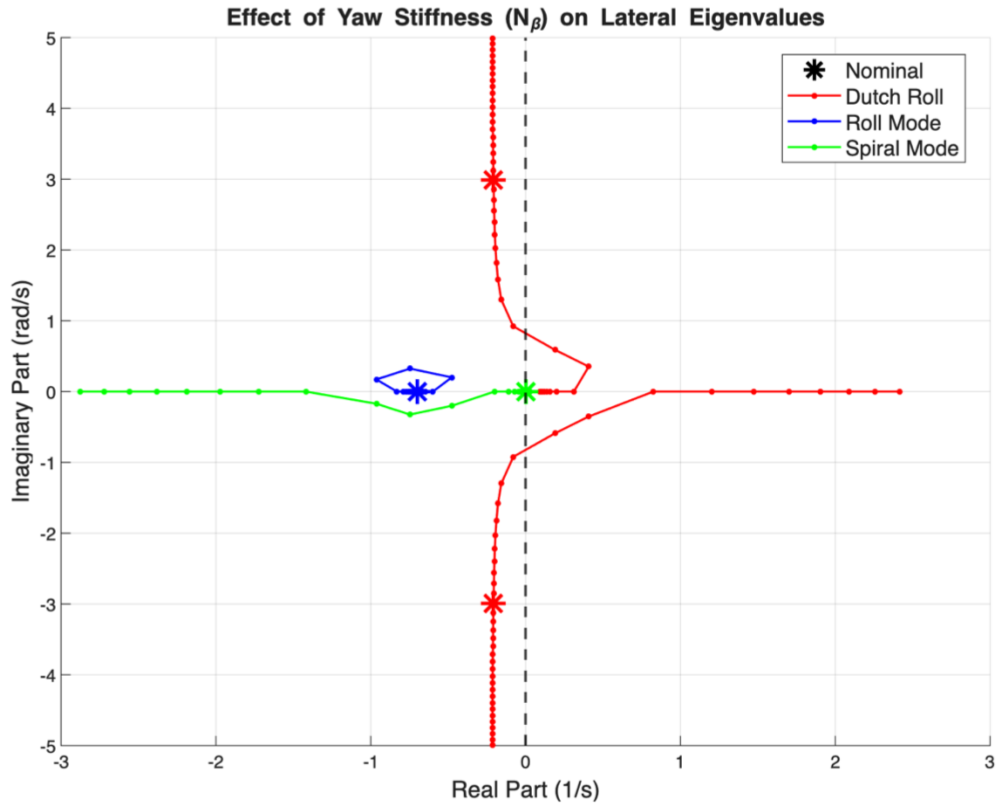


Effect of Roll Static Stability on Lateral Eigenvalues: At the nominal A-4D condition, the three lateral modes have eigenvalues of −0.207 ± 2.989j (Dutch Roll, $\zeta = 0.069$), −0.700 (Roll), and +0.003 (Spiral already mildly unstable). As $|\overline{L_\beta}|$ is increased from its nominal value of -14.24 ft/s²/rad by up to a factor of 3.0 (k = 20, increasing branch), the Spiral eigenvalue migrates leftward from +0.003 to -0.061, crossing the imaginary axis and becoming stable, while the Dutch Roll damping ratio simultaneously degrades from $\zeta = 0.069$ to $\zeta = 0.058$ and its real part shifts rightward from -0.207 to -0.192, a modest but consistent reduction in damping. The Roll mode eigenvalue moves only slightly, from -0.700 to -0.667, meaning that it is nearly insensitive to $\overline{L_\beta}$.

On the decreasing branch (k = 1 to 20, reducing $|\overline{L_\beta}|$), the Spiral eigenvalue moves further into the right-half plane, reaching +0.091 at f = -1.0, while the Dutch Roll recovers to $\zeta = 0.087$ with its real part shifting to -0.231. Notably, the Spiral is already marginally unstable at the nominal condition (+0.003), meaning any reduction in dihedral effect immediately worsens the divergence. The Roll mode remains confined between -0.667 and -0.741 across the entire sweep.

These results confirm the classical lateral-directional design tradeoff: increasing dihedral effect ($|\overline{L_\beta}|$) stabilizes the Spiral but at a direct cost to Dutch Roll damping, and for the A-4D the nominal configuration sits at a point where neither mode is fully satisfactory - the Spiral is marginally unstable and the Dutch Roll damping ratio of 0.069 is low.

6. YAW STIFFNESS ROOT LOCUS



Effect of Yaw Stiffness on Lateral Eigenvalues: At the nominal condition, the Dutch Roll mode has a natural frequency of ω = 2.997 rad/s and a damping ratio of ζ = 0.069, the Roll mode sits at -0.700, and the Spiral is marginally unstable at +0.003. As $\overline{N_\beta}$ is increased from its nominal value of 7.864 ft/s²/rad² by factors up to 3.0 (k = 20), the Dutch Roll natural frequency rises substantially from 2.997 to 4.998 rad/s while its real part stays nearly fixed between -0.207 and -0.214. Consequently, the damping ratio decreases from 0.069 to 0.043, meaning the mode becomes faster but proportionally less damped. The Roll mode eigenvalue barely moves, drifting only from -0.700 to -0.712 across the entire sweep, confirming its insensitivity to $\overline{N_\beta}$.

On the decreasing branch, as $\overline{N_\beta}$ is reduced toward zero, the Dutch Roll frequency decreases from 2.997 rad/s down to 0.926 rad/s at $\overline{N_\beta} = 0$, while its damping ratio peaks near ζ = 0.120 at f = 0.1 before the complex pair coalesces onto the real axis. For negative $\overline{N_\beta}$ (directionally unstable aircraft), the DR pair splits into two real roots: at f = -0.5 these are +1.473 and +0.158, both in the right-half plane, indicating a full static divergence in the coupled lateral-directional motion, while at f = -1.0 one root reaches +2.415. The Spiral mode shows only marginal sensitivity throughout the increasing branch, drifting from +0.003 to +0.029, remaining weakly unstable in both cases.

7. STEADY CLIMBING TURN - CONTROL DEFLECTIONS

phi_turn = 0.5646 rad (32.35 deg)

theta_turn = 0.3281 rad (18.80 deg)

n = 1.165715, g*(n-1) = 5.33179 ft/s^2

p = -0.011249, q = 0.017681, r = 0.027916 rad/s

Control deflections (changes from level-flight trim):

Elevator Delta_de = +0.015852 rad (+0.9082 deg)

Aileron   Delta_da = -0.001801 rad (-0.1032 deg)

Rudder   Delta_dr = -0.002226 rad (-0.1275 deg)

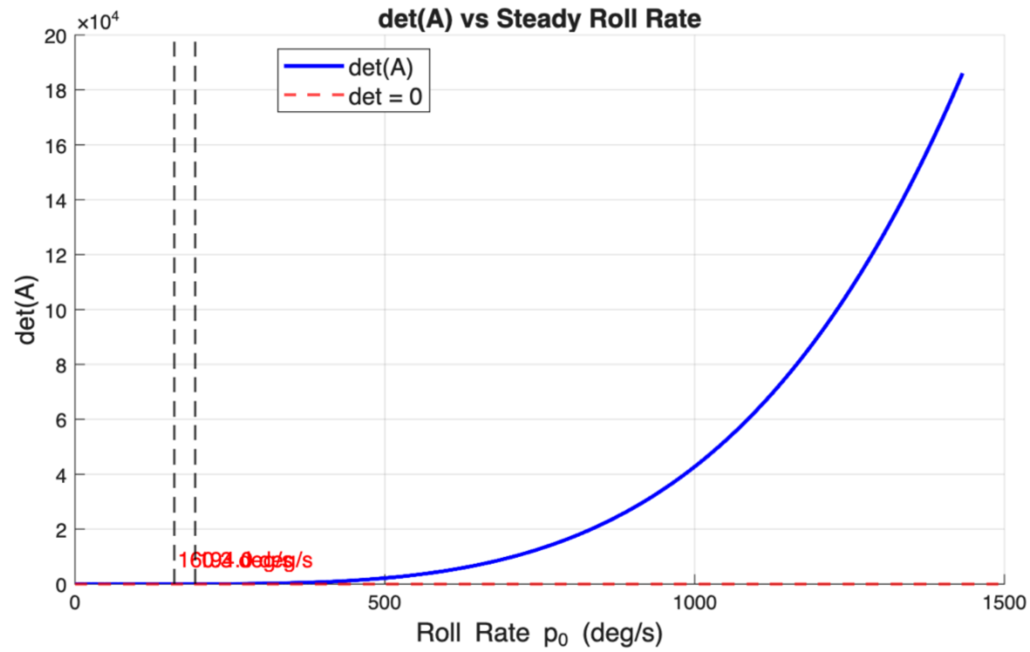
8. INERTIAL CROSS-COUPLING INSTABILITY CHECK

mu1 = (Izz-Ixx)/Iyy = 0.8073

mu2 = (Ixx-Iyy)/Izz = -0.6086

Aircraft IS prone to inertial cross-coupling instability.

Unstable range: 2.798 to 3.385 rad/s (160.3 to 194.0 deg/s)



The det(A) criterion reveals that the A-4D is indeed prone to inertial cross-coupling instability. At $p_0 = 0$ (no steady roll), det(A) is positive, meaning that the decoupled aircraft is statically stable. As the steady roll rate increases from zero, det(A) decreases monotonically and crosses zero at approximately $p_0 = 2.80$ rad/s (160.3 deg/s), marking the onset of static instability in the coupled pitch-yaw system. det(A) remains negative (statically unstable aircraft) until $p_0 = 3.39$ rad/s (194.0 deg/s), where it crosses back through zero and returns to positive values, hence restoring static stability. The instability window is, therefore, relatively narrow in roll rate space, spanning approximately 0.59 rad/s (33.7 deg/s). This behavior is a direct consequence of the A-4D's inertia distribution, where the large difference between I_{yy} and I_{zz} relative to I_{xx} (quantified by $\mu_1 = 0.8073$

and $\mu_2 = -0.6086$) causes the inertial coupling terms in the pitch and yaw moment equations to become destabilizing at sustained roll rates in this band. The practical implication is that the pilot must avoid sustaining roll rates in the range of 160 deg/s to 194 deg/s, as the aircraft would experience a divergent coupled pitching and yawing motion that cannot be arrested without reducing the roll rate below the lower instability boundary.

9. Appendix

```
clear; clc; close all;
%% ==========================================================
%  GIVEN DATA
%% ==========================================================
W = 17578; S = 260; b = 27.5; cbar = 10.8; g = 32.1745;
Ixx = 8190; Iyy = 25900; Izz = 29100; Ixz = -1952;
mass = W/g;
T_K = 218.92;                          % standard atmosphere temp at 35,000 ft, Kelvin
T_R = T_K * 9/5;                       % convert to Rankine (imperial)
a_sound = sqrt(1.4 * 1716 * T_R);      % speed of sound, ft/s (R_imperial = 1716 ft^2/s^2/R)
V0  = 0.6 * a_sound;                   % trim airspeed at M=0.6, ft/s
alpha_trim = 8.8*pi/180;  theta0 = alpha_trim;
cos_th = cos(theta0); sin_th = sin(theta0); tan_th = tan(theta0);
fprintf('V0 = %.4f ft/s,  alpha_trim = %.1f deg\n', V0, alpha_trim*180/pi);

% Normalized longitudinal derivatives
Xu_b=-0.0128; Xa_b=-20.61;  Xde_b= 2.69;
Zu_b=-0.114;  Za_b=-218.3;  Zde_b=-23.73;
Mu_b= 0.0004; Ma_b=-5.402;  Mad_b=-0.160; Mq_b=-0.484; Mde_b=-8.1;

% Normalized lateral derivatives
Yb_b=-60.38; Yda_b=-0.478; Ydr_b=10.46;
Lb_b=-14.24; Lp_b=-0.671; Lr_b= 0.464; Lda_b=7.998; Ldr_b=2.739;
Nb_b= 7.864; Np_b=-0.004; Nr_b=-0.291; Nda_b=-0.139; Ndr_b=-3.517;

% Inertia-coupled (primed) lateral derivatives
Gamma = 1 - Ixz^2/(Ixx*Izz);
Lb_p =(Lb_b +(Ixz/Ixx)*Nb_b)/Gamma;  Lp_p=(Lp_b+(Ixz/Ixx)*Np_b)/Gamma;
Lr_p =(Lr_b +(Ixz/Ixx)*Nr_b)/Gamma;  Lda_p=(Lda_b+(Ixz/Ixx)*Nda_b)/Gamma;
Ldr_p=(Ldr_b+(Ixz/Ixx)*Ndr_b)/Gamma;
Nb_p =(Nb_b +(Ixz/Izz)*Lb_b)/Gamma;  Np_p=(Np_b+(Ixz/Izz)*Lp_b)/Gamma;
Nr_p =(Nr_b +(Ixz/Izz)*Lr_b)/Gamma;  Nda_p=(Nda_b+(Ixz/Izz)*Lda_b)/Gamma;
Ndr_p=(Ndr_b+(Ixz/Izz)*Ldr_b)/Gamma;
```

Part (i): STATE-SPACE MATRICES

```
%% =========================================================
%  %  x_lon = [Du/V, Da, q, Dth]^T   u_lon = [Dde]
%  x_lat = [beta, p, r, phi]^T    u_lat = [Dda, Ddr]^T
%
%  A_lon and B_lon are obtained via similarity transform T=diag([V0,1,1,1]):
%    A_DuV = T^-1 * A_DV * T,   B_DuV = T^-1 * B_DV
%  where A_DV is the standard form with DeltaV (ft/s) as first state.
%% =========================================================

% Build standard form first (state: DeltaV in ft/s)
A_DV = [Xu_b,                       Xa_b,                0,            -g*cos_th          ;
        Zu_b/V0,                    Za_b/V0,             1,            -g*sin_th/V0       ;
        Mu_b+Mad_b*(Zu_b/V0),       Ma_b+Mad_b*(Za_b/V0),  Mq_b+Mad_b, -Mad_b*g*sin_th/V0 ;
        0,                          0,                   1,            0                  ];
B_DV = [Xde_b; Zde_b/V0; Mde_b+Mad_b*(Zde_b/V0); 0];

% Similarity transform to Du/V state: x1 = DeltaV/V0
T_scale = diag([V0, 1, 1, 1]);
A_lon = T_scale \ A_DV * T_scale;   % T^-1 * A_DV * T
B_lon = T_scale \ B_DV;             % T^-1 * B_DV

A_lat = [Yb_b/V0, 0,    -1,     g*cos_th/V0;
         Lb_p,    Lp_p,  Lr_p,  0          ;
         Nb_p,    Np_p,  Nr_p,  0          ;
         0,       1,     tan_th, 0         ];

B_lat = [Yda_b/V0, Ydr_b/V0;
         Lda_p,    Ldr_p   ;
         Nda_p,    Ndr_p   ;
         0,        0       ];

fprintf('\nLongitudinal A matrix:\n'); disp(A_lon);
fprintf('Longitudinal B vector:\n');  disp(B_lon);
fprintf('Lateral A matrix:\n');       disp(A_lat);
fprintf('Lateral B matrix:\n');       disp(B_lat);
```

Part (ii): EIGENVALUES, EIGENVECTORS, MODE IDENTIFICATION

```
[V_lon, D_lon] = eig(A_lon);  eigs_lon = diag(D_lon);
[V_lat, D_lat] = eig(A_lat);  eigs_lat = diag(D_lat);

T_lon = diag([1, 1, 2*V0/cbar, 1]);   % x1=Du/V already dimensionless; only q needs scaling
T_lat = diag([1,  2*V0/b, 2*V0/b, 1]);
V_lon_nd = T_lon \ V_lon;
V_lat_nd = T_lat \ V_lat;

norm_ev = @(v) v / v(abs(v)==max(abs(v)));

% Longitudinal: sort by |imag| descending (SP first, Phugoid second)
[~,idx] = sort(abs(imag(eigs_lon)),'descend');
eigs_lon = eigs_lon(idx);  V_lon_nd = V_lon_nd(:,idx);

lon_labels = {'Delta_u/V0','Dalpha','q*cbar/(2V0)','Dtheta'};
% Physical dominant states per Ch.9 characterization:
lon_dom    = {'Dalpha, q*cbar/(2V0)','Delta_u/V0, Dtheta'};
lon_names  = {'Short Period','Phugoid'};

fprintf('\n--- LONGITUDINAL MODES ---\n');
    for m = 1:2
        k = 2*m-1;  ev = eigs_lon(k);  evec = norm_ev(V_lon_nd(:,k));
        fprintf('\n%s:  lambda = %+.5f %+.5fi\n', lon_names{m}, real(ev), imag(ev));
        fprintf('  Non-dim eigenvector (magnitude, phase):\n');
        for j=1:4
            fprintf('    |%s| = %.4f,  phase = %.1f deg\n', ...
                    lon_labels{j}, abs(evec(j)), angle(evec(j))*180/pi);
        end
        fprintf('  Dominant states: %s\n', lon_dom{m});
        if real(ev)<0, fprintf('  Stability: Stable\n');
        else,          fprintf('  Stability: UNSTABLE\n'); end
    end

    % Lateral: separate DR (complex) from Roll/Spiral (real)
    is_cx    = abs(imag(eigs_lat)) > 1e-6;
    idx_DR   = find(is_cx & imag(eigs_lat)>0);
    idx_real = find(~is_cx);
    [~,srt]  = sort(abs(real(eigs_lat(idx_real))),'descend');
    idx_roll   = idx_real(srt(1));
    idx_spiral = idx_real(srt(2));

    lat_labels = {'beta','p*b/(2V0)','r*b/(2V0)','phi'};
```

```
    lat_idx    = [idx_DR, idx_roll, idx_spiral];
    lat_names  = {'Dutch Roll','Roll','Spiral'};

    fprintf('\n--- LATERAL MODES ---\n');
    for m = 1:3
        k = lat_idx(m);  ev = eigs_lat(k);  evec = norm_ev(V_lat_nd(:,k));
        if abs(imag(ev))>1e-6
            fprintf('\n%s:  lambda = %+.5f %+.5fi\n', lat_names{m}, real(ev), imag(ev));
        else
            fprintf('\n%s:  lambda = %+.5f (real)\n', lat_names{m}, real(ev));
        end
        fprintf('  Non-dim eigenvector (magnitude, phase):\n');
        for j=1:4
            if abs(imag(ev))>1e-6
                fprintf('    |%s| = %.4f,  phase = %.1f deg\n', ...
                        lat_labels{j}, abs(evec(j)), angle(evec(j))*180/pi);
            else
                fprintf('    |%s| = %.4f\n', lat_labels{j}, abs(evec(j)));
            end
        end
        [~,dom] = sort(abs(evec),'descend');
        if abs(imag(ev))>1e-6
            fprintf('  Dominant states: %s, %s\n', lat_labels{dom(1)}, lat_labels{dom(2)});
        else
            fprintf('  Dominant state: %s\n', lat_labels{dom(1)});
        end
        if real(ev)<0,           fprintf('  Stability: Stable\n');
        elseif abs(real(ev))<1e-6, fprintf('  Stability: Neutral\n');
        else,                    fprintf('  Stability: UNSTABLE\n'); end
    end

    fprintf('\n--- SUMMARY TABLE ---\n');
    fprintf('%-14s %-30s %s\n','Mode','Eigenvalue','Stability');
    fprintf('%s\n',repmat('-',1,55));
    all_ev    = [eigs_lon(1);eigs_lon(3);eigs_lat(idx_DR);eigs_lat(idx_roll);eigs_lat(idx_spiral)];
    all_names = {'Short Period','Phugoid','Dutch Roll','Roll','Spiral'};
    for k=1:5
        ev=all_ev(k);
```

```
        if abs(imag(ev))>1e-6, evs=sprintf('%+.5f%+.5fi',real(ev),imag(ev));
        else, evs=sprintf('%+.5f',real(ev)); end
        if real(ev)<0, st='Stable'; elseif abs(real(ev))<1e-6, st='Neutral'; else, st='UNSTABLE'; end
        fprintf('%-14s %-30s %s\n',all_names{k},evs,st);
    end
```

Part (iii): MODAL PARAMETERS

```
    eigs_r = eig(A_lon);
    cx = eigs_r(abs(imag(eigs_r))>1e-6 & imag(eigs_r)>0);
    [~,si]=sort(abs(cx),'descend');
    ev_SP=cx(si(1)); ev_PH=cx(si(2));

    eigs_rl = eig(A_lat);
    cx_l = eigs_rl(abs(imag(eigs_rl))>1e-6 & imag(eigs_rl)>0); ev_DR=cx_l(1);
    rl_l = real(eigs_rl(abs(imag(eigs_rl))<=1e-6));
    [~,sr]=sort(abs(rl_l),'descend');  ev_Roll=rl_l(sr(1)); ev_Spiral=rl_l(sr(2));

    fprintf('\n%-14s %9s %8s %8s %10s %14s\n','Mode','wn(r/s)','zeta','wd(r/s)','Period(s)','T_half/dbl(s)');
    fprintf('%s\n',repmat('-',1,68));
    for c = {{'Short Period',ev_SP},{'Phugoid',ev_PH},{'Dutch Roll',ev_DR}}
        nm=c{1}{1}; ev=c{1}{2};
        sig=real(ev); wd=abs(imag(ev)); wn=abs(ev);
        zeta=-sig/wn; T=2*pi/wd; Th=log(2)/(abs(zeta)*wn);
        if sig<0, lb='T_half'; else, lb='T_double'; end
        fprintf('%-14s %9.4f %8.4f %8.4f %10.4f %10.4f (%s)\n',nm,wn,zeta,wd,T,Th,lb);
    end
    fprintf('%s\n',repmat('-',1,68));

    fprintf('\n%-14s %12s %12s %14s\n','Mode','Eigenvalue','tau(s)','T_half/dbl(s)');
    fprintf('%s\n',repmat('-',1,55));
    for c = {{'Roll',ev_Roll},{'Spiral',ev_Spiral}}
        nm=c{1}{1}; ev=c{1}{2};
        tau=1/abs(ev); Th=log(2)/abs(ev);
        if ev<0, lb='T_half'; else, lb='T_double'; end
        fprintf('%-14s %+12.5f %12.4f %10.4f (%s)\n',nm,ev,tau,Th,lb);
    end
    fprintf('%s\n',repmat('-',1,55));
```

Part (iv): PITCH STIFFNESS ROOT LOCUS

```
factors_iv = unique([1+0.1*(0:20), 1-0.1*(0:20)]);
eigs_iv    = zeros(length(factors_iv),4);
for i=1:length(factors_iv)
    At=A_lon; At(3,2)=Ma_b*factors_iv(i)+Mad_b*(Za_b/V0);
    ev=eig(At); [~,si]=sort(real(ev),'ascend'); eigs_iv(i,:)=ev(si).';
end

figure('Name','Part iv - Pitch Stiffness','Position',[100 100 900 600]);
hold on; grid on;
ev_n=eig(A_lon);
h0=plot(real(ev_n),imag(ev_n),'k*','MarkerSize',14,'LineWidth',2,'DisplayName','Nominal');
h1=plot(real(eigs_iv(:,1:2)),imag(eigs_iv(:,1:2)),'b.-','MarkerSize',8,'DisplayName','Short Period');
set(h1(2),'HandleVisibility','off');
h2=plot(real(eigs_iv(:,3:4)),imag(eigs_iv(:,3:4)),'r.-','MarkerSize',8,'DisplayName','Phugoid');
set(h2(2),'HandleVisibility','off');
xline(0,'k--','LineWidth',1.5,'HandleVisibility','off');
xlabel('Real Part (1/s)','FontSize',13); ylabel('Imaginary Part (rad/s)','FontSize',13);
title('Effect of Pitch Stiffness (M_\alpha) on Longitudinal Eigenvalues','FontSize',13);
legend([h0,h1(1),h2(1)],{'Nominal','Short Period','Phugoid'},'FontSize',12,'Location','best');
text(-1.5,3.5,'Short Period','FontSize',11,'Color','b');
text(0.05,0.1,'Phugoid','FontSize',11,'Color','r');
hold off;
```

Part (v): ROLL STIFFNESS ROOT LOCUS

```
factors_v = unique([1+0.1*(1:20), 1-0.1*(1:20)]);
N_v = length(factors_v);
eigs_v = zeros(N_v,4);

% Initialize column order at nominal using |imag| sort + |real| sort for real pair
ev_init = eig(A_lat);
[~,si0] = sort(abs(imag(ev_init)),'descend'); ev_init = ev_init(si0);
```

```
    [~,si2] = sort(abs(real(ev_init(3:4))),'descend'); ev_init(3:4)=ev_init(si2+2);
    eigs_v(1,:) = ev_init.';

    % Re-compute first sweep point
    Lv=Lb_b*factors_v(1); At=A_lat;
    At(2,1)=(Lv+(Ixz/Ixx)*Nb_b)/Gamma; At(3,1)=(Nb_b+(Ixz/Izz)*Lv)/Gamma;
    evn=eig(At); prev=ev_init; asgn=false(4,1); ord=zeros(4,1);
    for c=1:4
        d=abs(evn-prev(c)); d(asgn)=inf; [~,b]=min(d); ord(c)=b; asgn(b)=true;
    end
    eigs_v(1,:)=evn(ord).';

    % Nearest-neighbour tracking across sweep
    for i=2:N_v
        Lv=Lb_b*factors_v(i); At=A_lat;
        At(2,1)=(Lv+(Ixz/Ixx)*Nb_b)/Gamma; At(3,1)=(Nb_b+(Ixz/Izz)*Lv)/Gamma;
        evn=eig(At); prev=eigs_v(i-1,:).'; asgn=false(4,1); ord=zeros(4,1);
        for c=1:4
            d=abs(evn-prev(c)); d(asgn)=inf; [~,b]=min(d); ord(c)=b; asgn(b)=true;
        end
        eigs_v(i,:)=evn(ord).';
    end

    ev_lat_n = eig(A_lat);
    figure('Name','Part v - Roll Stiffness','Position',[100 100 900 600]);
    hold on; grid on;
    hn=plot(real(ev_lat_n),imag(ev_lat_n),'k*','MarkerSize',14,'LineWidth',2,'DisplayName','Nominal');
    hd=plot(real(eigs_v(:,1:2)),imag(eigs_v(:,1:2)),'b.-','MarkerSize',8,'DisplayName','Dutch Roll');
    set(hd(2),'HandleVisibility','off');
    hr=plot(real(eigs_v(:,3)),imag(eigs_v(:,3)),'r.-','MarkerSize',8,'DisplayName','Roll');
    hs=plot(real(eigs_v(:,4)),imag(eigs_v(:,4)),'m.-','MarkerSize',8,'DisplayName','Spiral');
    xline(0,'k--','LineWidth',1.5,'HandleVisibility','off');
    xlabel('Real Part (1/s)','FontSize',13); ylabel('Imaginary Part (rad/s)','FontSize',13);
```

```
title('Effect of Roll Static Stability (L_\beta) on Lateral Eigenvalues','FontSize',13);
legend([hn,hd(1),hr,hs],{'Nominal','Dutch Roll','Roll','Spiral'},'FontSize',12,'Location','best');
hold off;
fprintf('  Effect of roll stiffness (L_bar_beta):\n');
fprintf('  Increasing |L_bar_beta|: spiral mode moves left (more stable); DR moves right (less stable damping).\n');
fprintf('  Decreasing |L_bar_beta|: spiral moves right and crosses imaginary axis (unstable spiral).\n');
fprintf('  Roll mode is nearly unaffected (Roll is governed primarily by L_bar_p, not L_bar_beta).\n');
```

Part (vi): YAW STIFFNESS ROOT LOCUS

```
k_vi            = 1:20;
factors_yaw_inc = 1 + 0.1*k_vi;
factors_yaw_dec = 1 - 0.1*k_vi;

ev0    = eig(A_lat);
is_cx  = abs(imag(ev0)) > 1e-6;
cx_ev  = ev0(is_cx);   rl_ev = ev0(~is_cx);
[~,si] = sort(imag(cx_ev),'descend');  cx_ev = cx_ev(si);
[~,si] = sort(real(rl_ev),'ascend');   rl_ev = rl_ev(si);
nom_ordered = [cx_ev; rl_ev];   % 4x1, consistent column order

% --- Track increasing branch ---
prev    = nom_ordered;
N       = length(k_vi);
eigs_yi = zeros(N+1, 4);
eigs_yi(1,:) = prev.';
for i = 1:N
    Nv = Nb_b * factors_yaw_inc(i);
    At = A_lat;
    At(2,1) = (Lb_b + (Ixz/Ixx)*Nv) / Gamma;
    At(3,1) = (Nv   + (Ixz/Izz)*Lb_b) / Gamma;
    evn = eig(At);
    asgn = false(4,1);  ord = zeros(4,1);
    for c = 1:4
        d = abs(evn - prev(c));  d(asgn) = inf;
        [~,b] = min(d);  ord(c) = b;  asgn(b) = true;
    end
    prev = evn(ord);
    eigs_yi(i+1,:) = prev.';
end
```

```
    % --- Track decreasing branch (reset prev to nominal) ---
    prev    = nom_ordered;
    eigs_yd = zeros(N+1, 4);
    eigs_yd(1,:) = prev.';
    for i = 1:N
        Nv = Nb_b * factors_yaw_dec(i);
        At = A_lat;
        At(2,1) = (Lb_b + (Ixz/Ixx)*Nv) / Gamma;
        At(3,1) = (Nv   + (Ixz/Izz)*Lb_b) / Gamma;
        evn = eig(At);
        asgn = false(4,1);  ord = zeros(4,1);
        for c = 1:4
            d = abs(evn - prev(c));  d(asgn) = inf;
            [~,b] = min(d);  ord(c) = b;  asgn(b) = true;
        end
        prev = evn(ord);
        eigs_yd(i+1,:) = prev.';
    end

    % --- Plot: each branch is a separate connected series ---
    ev_lat_nom = eig(A_lat);

    figure('Name','Part vi - Yaw Stiffness','Position',[100 100 900
650]);
    hold on; grid on;

    % Increasing branch
    plot(real(eigs_yi(:,1)), imag(eigs_yi(:,1)), 'r.-', 'MarkerSize',10,
'LineWidth',1.2, 'HandleVisibility','off');
    plot(real(eigs_yi(:,2)), imag(eigs_yi(:,2)), 'r.-', 'MarkerSize',10,
'LineWidth',1.2, 'HandleVisibility','off');
    plot(real(eigs_yi(:,3)), imag(eigs_yi(:,3)), 'b.-', 'MarkerSize',10,
'LineWidth',1.2, 'HandleVisibility','off');
    plot(real(eigs_yi(:,4)), imag(eigs_yi(:,4)), 'g.-', 'MarkerSize',10,
'LineWidth',1.2, 'HandleVisibility','off');

    % Decreasing branch
    plot(real(eigs_yd(:,1)), imag(eigs_yd(:,1)), 'r.-', 'MarkerSize',10,
'LineWidth',1.2, 'HandleVisibility','off');
    plot(real(eigs_yd(:,2)), imag(eigs_yd(:,2)), 'r.-', 'MarkerSize',10,
'LineWidth',1.2, 'HandleVisibility','off');
    plot(real(eigs_yd(:,3)), imag(eigs_yd(:,3)), 'b.-', 'MarkerSize',10,
'LineWidth',1.2, 'HandleVisibility','off');
    plot(real(eigs_yd(:,4)), imag(eigs_yd(:,4)), 'g.-', 'MarkerSize',10,
'LineWidth',1.2, 'HandleVisibility','off');
```

```
    % Nominal star markers
    plot(real(cx_ev), imag(cx_ev), 'r*', 'MarkerSize',14, 'LineWidth',2, 'HandleVisibility','off');
    plot(real(rl_ev(1)), 0, 'b*', 'MarkerSize',14, 'LineWidth',2, 'HandleVisibility','off');
    plot(real(rl_ev(2)), 0, 'g*', 'MarkerSize',14, 'LineWidth',2, 'HandleVisibility','off');

    % Dummy handles for legend only
    h_DR  = plot(NaN,NaN,'r.-',  'MarkerSize',10, 'LineWidth',1.2, 'DisplayName','Dutch Roll');
    h_Ro  = plot(NaN,NaN,'b.-',  'MarkerSize',10, 'LineWidth',1.2, 'DisplayName','Roll Mode');
    h_Sp  = plot(NaN,NaN,'g.-',  'MarkerSize',10, 'LineWidth',1.2, 'DisplayName','Spiral Mode');
    h_nom = plot(NaN,NaN,'k*',   'MarkerSize',12, 'LineWidth',2, 'DisplayName','Nominal');

    xline(0,'k--','LineWidth',1.5,'HandleVisibility','off');
    xlabel('Real Part (1/s)',        'FontSize',13);
    ylabel('Imaginary Part (rad/s)', 'FontSize',13);
    title('Effect of Yaw Stiffness (N_\beta) on Lateral Eigenvalues', 'FontSize',13);
    legend([h_nom, h_DR, h_Ro, h_Sp], 'FontSize',12, 'Location','best');
    hold off;

    fprintf('  Increasing Nb: DR frequency increases (moves up imaginary axis).\n');
    fprintf('  Decreasing Nb to 0: DR coalesces on real axis.\n');
    fprintf('  Negative Nb (directionally unstable): one DR root enters RHP.\n');
    fprintf('  Roll mode nearly insensitive to Nb changes.\n');
```

Part (vii): STEADY CLIMBING TURN — CONTROL DEFLECTIONS

```
    gamma_c = 10*pi/180;  psi_dot = 2*pi/180;
    phi_turn   = atan(V0*psi_dot/g);              % Eq.11.6
    theta_turn = alpha_trim + gamma_c;             % Eq.11.7
    n_load     = cos(gamma_c)/cos(phi_turn);       % load factor

    p_turn = -psi_dot*sin(theta_turn);
    q_turn =  psi_dot*sin(phi_turn)*cos(theta_turn);
    r_turn =  psi_dot*cos(phi_turn)*cos(theta_turn);

    fprintf('  phi_turn   = %.4f rad (%.2f deg)\n', phi_turn, phi_turn*180/pi);
```

```
    fprintf('  theta_turn = %.4f rad (%.2f deg)\n', theta_turn, theta_turn*180/pi);
    fprintf('  n = %.6f,  g*(n-1) = %.5f ft/s^2\n', n_load, g*(n_load-1));
    fprintf('  p = %.6f,  q = %.6f,  r = %.6f  rad/s\n', p_turn, q_turn, r_turn);

    % Longitudinal: Eqs.11.1-11.2 [Dalpha, Ddelta_e]
    Ma_eff  = Ma_b  + Mad_b*(Za_b/V0);
    Mde_eff = Mde_b + Mad_b*(Zde_b/V0);
    Mq_eff  = Mq_b  + Mad_b;
    sol_lon   = [Za_b, Zde_b; Ma_eff, Mde_eff] \ [g*(n_load-1); -Mq_eff*q_turn];
    Delta_alpha = sol_lon(1);  Delta_de = sol_lon(2);

    % Lateral: Eqs.11.4-11.5, beta=0 [Ddelta_a, Ddelta_r]
    sol_lat = [Lda_p, Ldr_p; Nda_p, Ndr_p] \ ...
              [-Lr_p*r_turn - Lp_p*p_turn; -Nr_p*r_turn - Np_p*p_turn];
    Delta_da = sol_lat(1);  Delta_dr = sol_lat(2);

    fprintf('\nControl deflections (changes from level-flight trim):\n');
    fprintf('  Elevator  Delta_de = %+.6f rad  (%+.4f deg)\n', Delta_de, Delta_de*180/pi);
    fprintf('  Aileron   Delta_da = %+.6f rad  (%+.4f deg)\n', Delta_da, Delta_da*180/pi);
    fprintf('  Rudder    Delta_dr = %+.6f rad  (%+.4f deg)\n', Delta_dr, Delta_dr*180/pi);
```

Part (viii): INERTIAL CROSS-COUPLING INSTABILITY CHECK

```
    %% =========================================================
    %  Ch.12 Eq.(7): det(A(p0))=0 marks static instability boundary
    %  Coupled state: [Dalpha, q, beta, r]^T
    %% =========================================================

    mu1  = (Izz-Ixx)/Iyy;  mu2 = (Ixx-Iyy)/Izz;
    Zw_m = Za_b/V0;         Yv_m = Yb_b/V0;
    Ma_p = Ma_b+Mad_b*(Za_b/V0);   Mq_p = Mq_b+Mad_b;

    fprintf('  mu1 = (Izz-Ixx)/Iyy = %.4f\n', mu1);
    fprintf('  mu2 = (Ixx-Iyy)/Izz = %.4f\n', mu2);

    p0_vec = linspace(0, 25, 5000);
    det_v  = zeros(size(p0_vec));
    for i = 1:length(p0_vec)
```

```
        p0 = p0_vec(i);
        Acc = [Zw_m, 1,        -p0,       0      ;
               Ma_p, Mq_p, -Mad_b*p0,  p0*mu1  ;
               p0,   0,         Yv_m,      -1      ;
               0,    p0*mu2,    Nb_b,      Nr_b    ];
        det_v(i) = det(Acc);
    end

    ci = find(diff(sign(det_v))~=0);
    p0_cr = zeros(1,length(ci));
    for k=1:length(ci)
        i1=ci(k); i2=i1+1;
        p0_cr(k)=p0_vec(i1)-det_v(i1)*(p0_vec(i2)-p0_vec(i1))/(det_v(i2)-det_v(i1));
    end

    if isempty(p0_cr)
        fprintf('  Aircraft is NOT prone to inertial cross-coupling instability.\n');
        fprintf('  det(A) > 0 for all p0 in [0, 25] rad/s (0 to %.0f deg/s).\n', 25*180/pi);
        fprintf('  No roll rate range exists for which a real eigenvalue crosses zero.\n');
    else
        fprintf('  Aircraft IS prone to inertial cross-coupling instability.\n');
        for k=1:2:length(p0_cr)-1
            fprintf('  Unstable range: %.3f to %.3f rad/s  (%.1f to %.1f deg/s)\n',...
p0_cr(k),p0_cr(k+1),p0_cr(k)*180/pi,p0_cr(k+1)*180/pi);
        end
    end

    % --- Plot: det(A) vs p0 ---
    figure('Name','Part viii - det(A)','Position',[100 100 800 450]);
    hold on; grid on;
    plot(p0_vec*180/pi, det_v,'b-','LineWidth',2,'DisplayName','det(A)');
    yline(0,'r--','LineWidth',1.5,'DisplayName','det = 0');
    if ~isempty(p0_cr)
        for k=1:length(p0_cr)
            xline(p0_cr(k)*180/pi,'k--','LineWidth',1.2,'HandleVisibility','off');
            text(p0_cr(k)*180/pi+5,max(det_v)*0.05,sprintf('%.1f deg/s',p0_cr(k)*180/pi),...
```

```
                 'FontSize',11,'Color','r');
        end
    end
    xlabel('Roll Rate p_0 (deg/s)','FontSize',13);
    ylabel('det(A)','FontSize',13);
    title('det(A) vs Steady Roll Rate','FontSize',13);
    legend('FontSize',12,'Location','best');
    hold off;
```